Shayaan Uddin

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Research Experience

Institute for Protein Design, Baker Lab Mar. 2026 – Present
Undergraduate Researcher *Seattle, WA*

- Worked on post-training an all-atom protein design model with **GRPO** and **DGPO preference optimization**, using reward-guided beam search to find high-reward designs that are fed back into training.
- Ran evaluations and ablations behind enzyme-design pass rates rising from 20/18/11% to **89/74/72%** across three atomic-motif targets at < 0.6 Å backbone RMSD, while preserving structural diversity across 40+ clusters; binder interface energy improved from +1.86 to −12.02 REU on four held-out ligands.
- Ran and debugged distributed post-training experiments (32-sample rollout groups across 8 GPUs), diagnosing numerical instability, NaN advantages, low reward variance, diversity collapse, and protein-structure processing failures.
- Evaluated designs with **PyRosetta** separated-interface energy, **RosettaFold3** min-iPAE after refolding, backbone and ligand RMSD, clash and chain-break checks, and TM-score structural diversity.

Algoverse Jan. 2025 – Aug. 2025
Interpretability Researcher *Remote*

- Developed **Reward-Informed Sparse Autoencoders** to identify latent features associated with reasoning behavior in Gemma 2-2B and Llama 3.1-8B.
- Applied **GRPO** and chain-of-thought fine-tuning, achieving 0.83 cross-validation AUC and 0.75 held-out AUC on latent reasoning classification.

Selected Projects

When Do Protein Designs Commit? | *PyTorch, Flow Matching, AlphaFold2* Paper

- Built a branching framework for **Proteína-Complexa** that pauses generation at intermediate denoising steps and independently continues each state to measure when final design properties become determined.
- Analyzed **19 protein targets** across fold identity, pLDDT, designability, and interface quality; observed **14 distinct settling orders**, showing that design properties do not emerge in a universal sequence across targets.
- Trained lineage-blocked linear readouts on intermediate denoiser activations; internal activations predicted final outcomes better than matched decoded partial structures across all four design qualities.

ProteinClaw: Agentic Protein Binder Design | *Python, MCP, Codex, GEPA, RFdiffusion3* GitHub

- Built **ProteinClaw**, a Codex plugin exposing RFdiffusion3, ProteinMPNN, ESMFold, AlphaFold2-multimer, and BoltzGen as **MCP** tools, letting an agent run a binder-design campaign from target retrieval through interface scoring and reporting.
- Generated and evaluated hundreds of nanobody designs against GPCRs, including the drug-binding face of the **μ-opioid receptor** (MOR) in its active state, narrowing to five candidates passing structural and interface filters.
- Applied **GEPA** to evolve the agent's prompts, planning, and tool-use policies from multi-objective design feedback, with a teacher–student loop refining student generations into Pareto-improved preference pairs.

Education

University of Washington Sept. 2025 – Jun. 2028
Bachelor of Science in Computer Science *Seattle, WA*

Publications & Presentations

When Do Protein Designs Commit? Interpreting Intermediate States During Generation — First author @ NeurIPS 2026 Interp4Discovery Workshop

Online Search-Augmented Reinforcement Post-Training for All-Atom Protein Design — Second author, submitted to ICLR 2027

RI-SAEs Reveal Evolving Patterns of Latent Reasoning in LLMs — Accepted research poster, NEMI 2025

Technical Skills

Machine Learning: PyTorch, flow matching, diffusion models, reinforcement learning, GRPO, PPO, SFT, preference learning, sparse autoencoders
Protein Modeling: RFdiffusion3, ProteinMPNN, AlphaFold2-Multimer, Proteina-Complexa, ESMFold, PyRosetta
Systems: Python, Linux, Git, Slurm, Docker, multi-GPU experimentation, DDP, HPC workflows